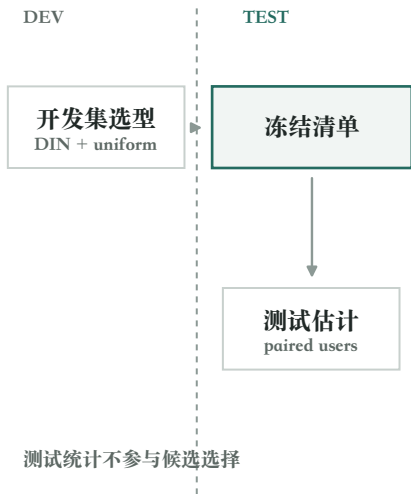
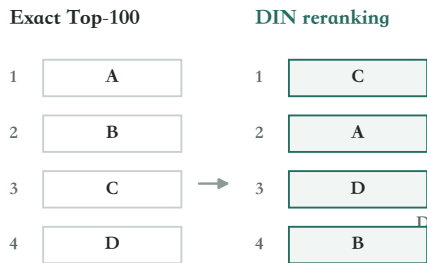


(a) 选择与冻结边界



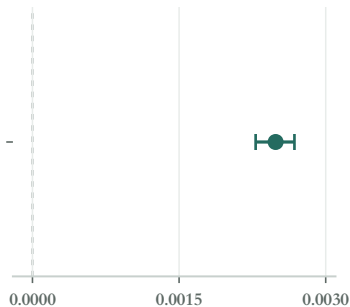
(b) 相同候选，重排前后



示意排序；候选集合固定

NDCG@100 0.02103 → 0.02351

(c) 用户级配对差值



95% CI [0.00228, 0.00268]

N = 50,653 配对用户

百分位 bootstrap · 单位=用户 · B=2,000

开发集选型 → 冻结清单 → 测试集用户级配对估计